

MANCHALA VENKATA DHARANI

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Portfolio — LinkedIn — GitHub — LeetCode

PROFESSIONAL SUMMARY

Computer Science undergraduate specializing in AI/ML, NLP, and Full-Stack Development with hands-on experience building intelligent web applications and transformer-based deep learning models. Skilled in Python, Machine Learning, and scalable backend development with strong foundations in DSA, DBMS, OS, and Computer Networks.

EDUCATION

B.Tech CSE , RGUKT Nuzvid	8.9/10
Intermediate , RGUKT Nuzvid	9.9/10
SSC , Vikas Vidhya Vihar EM School	599/600

TECHNICAL SKILLS

Languages: Python, C, SQL

Machine Learning: Classification, Regression, Feature Engineering, Model Evaluation

DL/NLP: CNN, RNN, LSTM, GRU, Transformers, BERT, NLP, TF-IDF, Sentiment Analysis

Libraries/Tools: Pandas, NumPy, Scikit-Learn, Git, GitHub, VS Code

Web/DB: HTML, CSS, JavaScript, Node.js, Flask, FastAPI, MongoDB

Core CS: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks

PROJECTS

AI-Based Communication Intelligence Platform

Live Demo — GitHub

Tech Stack: Python, FastAPI, NLP, Scikit-Learn, HTML, CSS

- Engineered AI-powered communication analysis platform evaluating 5+ speech parameters using NLP and ML techniques.
- Built real-time speech-to-text and automated feedback generation modules reducing manual evaluation effort by 60%.
- Created analytics dashboards and automated PDF reports for generating communication insights within 5 seconds.
- Integrated scalable full-stack architecture supporting real-time interaction and assessment workflows.

Transformer-Based Models for Stance Detection

Paper — GitHub

Tech Stack: Python, Transformers, NLP, BERT, BART, LSTM, GRU, GloVe

- Compared 8+ transformer and deep learning architectures including BERT, BART, LSTM, and GRU for stance detection.
- Trained and evaluated models on 5 benchmark datasets including P-Stance, VAST, COVID-19, and SemEval datasets.
- Achieved up to 80.7% accuracy using BART models with transformer attention mechanisms and contextual embeddings.
- Analyzed model performance using Accuracy, Precision, Recall, and F1-score metrics across multiple NLP tasks.

MindEase – Mental Health Stress Prediction Platform

Live Demo — GitHub

Tech Stack: Flask, MongoDB, Python, HTML, CSS, JavaScript

- Built full-stack mental health assessment and stress prediction platform with secure authentication and 10+ questionnaire-based evaluation features.
- Developed ML-driven stress prediction modules and wellness dashboards for behavioral stress analysis.
- Architected responsive frontend interfaces and integrated backend services for seamless user interaction.
- Optimized user engagement using predictive analytics and interactive dashboard visualization techniques.

ACHIEVEMENTS

- Qualified GATE 2026 in Computer Science and Engineering.
- Solved 120+ coding problems on LeetCode focused on Data Structures and Algorithms.
- Participated in hackathons and collaborative AI/ML development projects.

CERTIFICATIONS

NPTEL – Foundations of Deep Learning (Elite)

Deloitte Australia – Data Analytics Job Simulation

GeeksforGeeks – Python Programming Bootcamp

GeeksforGeeks – Full Stack Development Bootcamp